Engineering Chemistry (UE26CY151A) - Unit 1 Study Pack

Semester 1 · Compiled from the 2026 class slide text (decks 003-010) and the Unit 1 video transcripts. Generated 13 Oct 2026.

Status: PARTIAL FIRST PASS. This version covers the content that could be verified from slide text and the brief within the time box. It does *not* yet contain: rendered equations checked against slide images (the slide text omits every equation), decks 001, 002, 004 equations, 011, 012, the Raman material, the ESA ranking, the worked numerical examples, the separate documents 01-05, or the PDF versions. Formulas below are taken from the brief and marked as Not visually verified where the slide text does not show them.

Legend: Not on current slides · Not visually verified · [S0xx] = slide deck number, [T0xx mm:ss] = transcript time.

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- Not yet extracted in this pass

T3 Born-Oppenheimer approximation

Key points

- Total molecular energy = translational + rotational + vibrational + electronic energies [S003]
- Translational energy is negligibly small, so $E = E_{rot} + E_{vib} + E_{elec}$ [S003]
- Electronic, vibrational and rotational energies are treated as independent, because electronic levels are much farther apart than vibrational levels, which are much farther apart than rotational levels: $E_{rot} \ll E_{vib} \ll E_{elec}$ [S003]
- Transition energy map: rotational transitions are microwave, vibrational are infrared, electronic are optical or UV [S003, S007]

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E = E_trans + E_rot + E_vib + E_elec   (slide 003, equation not in text; standard form)
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T4 Rotational (microwave) spectroscopy

Key points

- Gross selection rule: molecule must have a **permanent dipole moment**. HCl and CO are microwave active; H_2, Cl_2 and CO_2 are not [S004 slides 3, 5]
- Model: rigid rotor for a heteronuclear diatomic, masses m_1, m_2 joined by a rod of length $r_0 = r_1 + r_2$, rotating about the centre of mass [S004 slide 6]
- Rotational quantum number J = 0, 1, 2, 3 … (integers from zero upwards) [S004 slide 8]
- Selection rule: $\Delta J = \pm 1$ [S004 slide 10]
- Rotational energy levels in cm^{-1}, with rotational constant B: $\varepsilon_J = BJ(J+1)$. Levels: J=0 gives 0, J=1 gives 2B, J=2 gives 6B, J=3 gives 12B [S004 slide 9]
- Line positions: the transition J → J+1 absorbs at $\Delta\varepsilon = 2B(J+1)$. The first line is at 2B; spacing between lines is constant at 2B: 2B, 4B, 6B, 8B, 10B [S004 slide 10-11]
- Information from the spectrum: B from the line spacing, then $I = \mu r_0^2$ gives the bond length r_0 [S004 final slide]
- Higher rotational levels are already populated at room temperature (intensity pattern in the spectrum) [S004]

> $\varepsilon_J = BJ(J+1)\ cm^{-1}$ Not visually verified (slide 8 equation missing in text)

> $I = \mu\ r_0^2, \quad \mu = m_1m_2/(m_1+m_2)$ Not visually verified (slides 7 and final slide)

> $\Delta\varepsilon(J \to J+1) = 2B(J+1)\ cm^{-1}$ Not visually verified

Exam trap: the first rotational line is at 2B, not B. J starts at 0.

T5 Vibrational spectroscopy: simple harmonic oscillator (SHO)

Key points

- Gross selection rule: the vibration must change the dipole moment (IR active) [S005]
- CO_2: symmetric stretch is IR inactive (no dipole change); asymmetric stretch and bending are IR active [S005]
- Bond modelled as a spring obeying Hooke's law; the potential energy curve is parabolic [S005]
- Energy levels are equally spaced; zero-point energy ε_0 is non-zero [S005]
- Selection rule $\Delta v = \pm 1$; so only **one line** is seen in the SHO spectrum [S005]
- From the spectrum the oscillation wavenumber is read, and the force constant k (bond strength) follows from it [S005]
- Every bond has a unique natural frequency, so each molecule's IR spectrum is unique (a fingerprint) [S005]

$\nu_{osc} = (1/2\pi)\sqrt{(k/\mu)}$ Hz Not visually verified (slide 005 equation missing in text)

$\varepsilon_v = (v + ½)\,\bar{\nu}_{osc}$ cm^{-1}, $\varepsilon_0 = ½\bar{\nu}_{osc}$ Not visually verified

T6 Vibrational spectroscopy: anharmonic oscillator

Key points

- Real bonds are anharmonic: stretched far enough, the bond breaks and the molecule dissociates [S006]
- Morse potential: PE = $D_e[1 - \exp\{a(r_0 - r)\}]^2$, where D_e is the depth of the PE curve and a is a constant [S006]
- Anharmonicity constant x_e is small and positive for bond stretching [S006]
- Energy spacing decreases as v increases; levels converge toward the dissociation limit [S006]
- Selection rule: Δv = ±1, ±2, ±3 … (overtones allowed) [S006]
- Most transitions start from v=0 (most populated). Fundamental: 0→1; first overtone: 0→2; second overtone: 0→3 [S006]
- Hot bands: at higher temperature, transitions from v=1 are also seen; the 1→2 hot band lies to the **left** of the fundamental [S006]

> $\varepsilon_v = (v+½)\bar{\omega}_e - (v+½)^2\bar{\omega}_e x_e$ cm^{-1} Not visually verified

> Fundamental = $\bar{\omega}_e(1 - 2x_e)$; First overtone = $2\bar{\omega}_e(1 - 3x_e)$ Not visually verified
> (from brief table, check against deck 006 slides 7 and 9)

T7 Electronic spectroscopy and the Franck-Condon principle

Key points

- HOMO → LUMO transitions in the UV-Visible region: σ→σ*, π→π*, n→&pi* [S007]
- All molecules can undergo electronic transitions (unlike IR or microwave) [S007]
- Electronic transitions carry vibrational structure (coarse structure) and rotational structure (fine structure) [S007]
- No selection rule restricts the vibrational level reached during an electronic transition [S007]
- Most transitions start from v″=0 (most populated). Lines are labelled (v′, v″); the set of lines is a progression; lines converge at high energy, showing anharmonicity in the excited state [S007]
- **Franck-Condon principle** (slide wording): “An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition.” Electronic transitions are therefore vertical [S007]
- Three cases [S007]:
  - r_e'' = r_e''': strongest line is (0,0); intensity falls for the rest
  - r_e'' > r_e''' (a little greater): strongest line is (v′, 0)
  - r_e'' » r_e''' (considerably greater): the molecule passes the dissociation energy, giving a continuum (kinetic energy of separated atoms is not quantised)
- Intensity is highest from the centre of v‴=0, where the probability of finding the nuclei is maximal [S007]

Note on the transcript: the class 007 transcript spells the principle "Frank Condon"; the correct name is Franck-Condon.

T8 Phase, component, degrees of freedom and the phase rule

Definitions

- **Phase:** any homogeneous, physically distinct part of a system, bounded by a surface and mechanically separable. Gas: P = 1 (gases are completely miscible). Liquid: P = number of layers if immiscible, 1 if miscible. Solid: each solid is a separate phase; each polymorph is a separate phase; P = 1 for a solid solution [S008]
- **Component:** the smallest number of independently varying chemical constituents needed to express the composition of every phase [S008]
- **Degrees of freedom (variance) F:** the minimum number of intensive variables (T, P, concentration) that must be fixed to define the system, equivalently the number that can vary without changing the number of phases [S008]
- Chemical potential: phases in equilibrium have equal chemical potential for each component [S009]

Counting examples

System	P	C	Reason
Ice / water / vapour (one pure substance)	3	1	Ice pieces still count as one phase
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, closed	3	2	3 species but any 2 of them define the composition
$NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$, closed	2	1	Gas composition fixed by NH_4Cl; C becomes 2 if extra HCl or NH_3 is added
$Na_2SO_4 \cdot 10H_2O$ salt hydrate	–	2	All phases expressed through Na_2SO_4 and H_2O

Statement of the phase rule

> `F = C − P + 2   (Gibbs, 1874; valid when equilibrium is not influenced by gravity, electric or magnetic fields)` [S008]

T9 Derivation of the phase rule; one-component water diagram

Derivation (slide 009)

- Variables: each of the P phases needs T, P and (C−1) mole fractions, so total variables = P(C−1) + 2 [S009]
- Equilibrium constraints: chemical potential of each component equal across the phases, giving (P−1) equations per component, so C(P−1) in total [S009]
- F = variables − constraints = P(C−1) + 2 − C(P−1) = C − P + 2 [S009]

One-component system (water)

- F = 3 − P [S009]
- One phase: F = 2, an **area** [S009]
- Two phases: F = 1, a **line** [S009]
- Three phases: F = 0, a **point** (triple point; for water 0.0098 °C and 4.58 mm Hg, as stated on slide 008) [S008, S009]
- High pressure, low temperature: solid. High temperature, low pressure: vapour [S009]

> `F = C − P + 2,  F = 3 − P (one component)` Not visually verified `for the constraint-count line`

Curves OA (vaporisation), OB (sublimation), OC (melting, negative slope) and the metastable OA′ from the water diagram are Not visually verified: the labels are in the diagram image, which the text extraction does not give. Render deck 009 before using this section in an exam.

T11 Reduced phase rule and the Pb-Ag eutectic system

Key points (slide 010)

- For two components with the full rule, F = 4 − P; at minimum P = 1, F = 3, which needs 3-D space, so it cannot be drawn on paper [S010]
- Measurements are usually at constant (atmospheric) pressure, which removes one degree of freedom: **F = C − P + 1**, the reduced (condensed) phase rule. Valid for condensed systems with no gas phase [S010]
- Pb–Ag: liquid miscible in all proportions; solid completely immiscible [S010]
- Melting points: Pb 327 °C, Ag 961 °C; eutectic temperature 303 °C at 2.6 % Ag (97.4 % Pb) [S010]
- Cooling path: first solid Pb forms at the Pb freezing curve; the melt becomes richer in Ag and moves along curve BO. At point O, solid Ag also separates; three phases (solid Ag, solid Pb, melt) coexist, F = 0 [S010]
- Areas: above BOA liquid melt; AOD solid Ag + liquid; BOC solid Pb + liquid; below COD solid Ag + solid Pb [S010]

Labelling conflict: the slide names A = 100 % Pb and B = 100 % Ag, but its text also describes curve BO as the freezing curve of Pb. Use the lettering your faculty uses in exams.

Not yet extracted in this pass

- **T1** EM radiation and matter, **T2** EM spectrum regions and Raman (deck 001, 002 equations not yet read) — use the 001 and 002 transcripts
- **T10** water diagram values and Clausius–Clapeyron reasoning (needs the 009 Part II transcript)
- **T12** cooling curves (deck 012, no video)
- **T13** computational chemistry (deck 011 and the Extra Topics PDF)
- Rendered slide images for equations (brief section 3.1), ESA past-paper ranking (section 5), worked numerical examples (section 5.1), symbol glossary, figures G1–G26, flowcharts F1–F14, the five themed documents 01–05, and PDF output.
- **Not on current slides, flagged:** Beer–Lambert law ($A = \log(I_0/I) = \varepsilon cl$) Not on current slides; Fe–C phase diagram Not on current slides; CO_2 phase diagram Not on current slides.

Missing video: class 003 (Vimeo 748278627) failed to download, HTTP 401. Class 012 never had a video.